

Class Meeting 12 Handout – Comments

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This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*. Diagrams were edited in *Affinity Publisher*

The handout features several tables of data. In this notebook we plot the data in tables 1 and 2 from exercise #1 of the class handout as we explore how Hammett σ values are defined. Then we will plot the data from tables 4, 5, 6 & 7 from exercise #2 of the handout. You can use the Jupyter notebooks accessible via the course web book to make your own versions of these plots.

Exercise: The Hammett Plot – 20 minutes

We are using the Hammett plot. Let us first consider the first simple exercise presented in the handout. Table 1 in the handout presents the pK_a values for a set of substituted benzoic acids. We observed that electron-withdrawing substituents decreased the pK_a values (more acidic) and electron-donating substituents increased the pK_a value (less acidic), as expected.

Table 2 presents a set of rate data for the base-catalyzed hydrolysis of ethyl esters of substituted benzoic acids. The rates changes with different substituents. Is the rate change similar to the change in pK_a ? Is there a correlation?

To test for a correlation we can plot the two values against each other and see if there is a trend. pK_a is a logarithmic value, so we will use the logarithm of the rate constants for the hydrolysis reaction. Let us plot $\log k$ vs pK_a as shown in figure 1

Table 1: Collated data from tables 1 & 2 from the class handout.

Substituent	pK_a	$\log \frac{k_X}{k_H}$
p-NO ₂	3.45	2.049
p-CN	3.56	1.729
p-Cl	4.0	0.851
p-H	4.20	0
p-CH ₃	4.36	-0.235
p-OCH ₃	4.47	-0.316

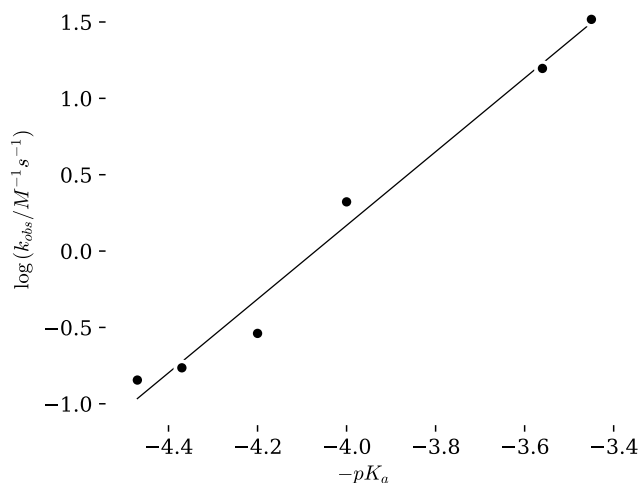


Figure 1: log-log plot of the rate constant for base hydrolysis of substituted ethylbenzoate vs the K_a for the corresponding benzoic acid.

The slope is $m = 2.4$.

This plot and the others generated from data sets in tables 1 & 2 can be investigated by using the *Python* notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table1and2_ClassExamples.ipynb.

It is linear and the slope is 2.4. The base hydrolysis reaction increases in rate to a greater degree than the change in pK_a . Another way to say this is that for every 10-fold increase in K_a there is a $10^{2.4}$ -fold increase in the rate constant of hydrolysis.¹

¹ Take note that the term $-pK_a$ is equivalent to $\log K_a$

The L-Word

LFER is the acronym for linear free-energy relationships. We have just seen that the logarithm of the rate of ester hydrolysis in base has a linear correlation with the $\log K_a$ value as represented by $-pK_a$ in figure 1 on the previous page.

Let us examine the equations for equilibrium and rate as a function of free-energy difference. In eq. 1 is the equation for equilibrium and in eq. 2 we present the Eyring equation for rate constants. We see that both have an exponential relationship to ΔG .

$$K_{eq} = e^{\frac{-\Delta G^\circ}{RT}} \quad (1)$$

$$k = \frac{\kappa k_B T}{h} e^{\frac{-\Delta G^\ddagger}{RT}} \quad (2)$$

If the value of $\Delta G^\ddagger$ for hydrolysis was changed 2.4-fold more than the value of ΔG° for acid equilibrium then we would see that slope of 2.4 in the log-log plot.

A proportional change in ΔG between two chemical systems will result in a linear log-log plot with a slope equivalent to the proportionality constant. The change in ΔG° for acid equilibrium of substituted benzoic acids should be proportional to the change in $\Delta G^\ddagger$ for reactions that feature a substituted benzene ring near the reaction centre.²

What's the Score?

We just observed a linear free-energy relationship between two related reactions: the acid equilibrium of substituted benzoic acids and the rate of base hydrolysis of substituted benzoic acid esters. Rather than use the acid equilibrium as the x -axis we could establish a more universal scoring system. Hammett defined the “Hammett parameter”, σ , to be a score for the electron density of a substituted benzene ring. It is now generally used as a score for the electron donating or withdrawing nature of a substituent group. The σ value started as the value of $\log(K_{aX}/K_{aH})$. This is the same as stating that σ is the difference between the pK_a values of substituted benzoic acids and the value for the unsubstituted case as shown in eq. 3.

$$\begin{aligned} \sigma &= \log \frac{K_{aX}}{K_{aH}} \\ \sigma &= \log K_{aX} - \log K_{aH} \\ \sigma &= -pK_{aX} + pK_{aH} \\ \sigma &= pK_{aH} - pK_{aX} \\ \sigma &= \Delta pK_a \end{aligned} \quad (3)$$

² Fine print: The effects of the substituent must be the same in both cases. Hammett chose to use only *m*- and *p*-substituted benzene rings so that the reaction centre is not affected by the steric bulk of the substituent. Acid equilibrium has a very small steric demand compared to nucleophilic addition at a carbonyl group so change in ΔG due to sterics would not be proportional between the two systems. If *ortho* substituents are not used then only the electronic effects will change ΔG . Hammett's system is thus limited to using substituted benzene rings as the probe for electronic effects in reactions and equilibrium.

Over time, many contributors have determined the effect of various substituents on the pK_a of benzoic acid and related that to the change in ΔG of a given reaction. After thousands of reactions were evaluated, chemists began to tweak the values so that the best overall fitting score could be achieved across this enormous database. Modern σ values remain very close or identical to the ΔpK_a of the corresponding substituted benzoic acid. This has also resulted in slightly different sets of σ values depending on the database used for optimization. When there are discrepancies between two tables of *sigma* values in the literature, the differences are usually very small and unimportant. If there is a large discrepancy then there is likely a typographical error in one of the tables.³

³ Typographical errors are common. Always validate your data. Blaming the literature after the Mars probe crashes won't change the fact that billions of dollars was wasted. Professionals stand by their work, they don't blame their data.

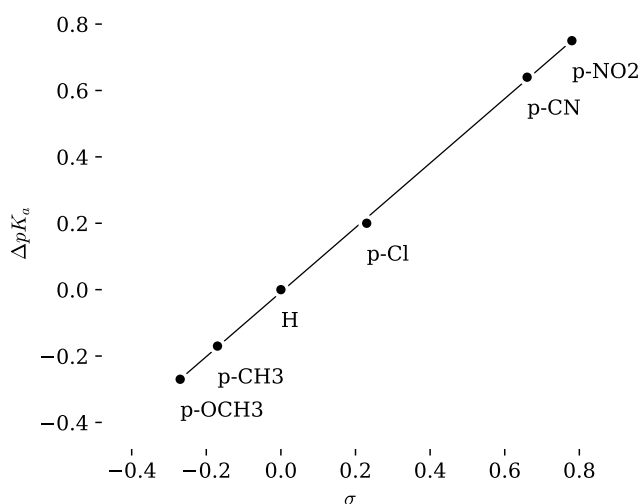


Figure 2: Plot of ΔpK_a for substituted benzoic acids vs the literature σ values for the corresponding substituents. Observe that there is almost perfect equivalence. Almost, the σ values have been adjusted over time to give the best overall agreement with experimental observations in a wide range of systems. They are originated as the ΔpK_a of substituted benzoic acids but have evolved since then.

Now we have a score for the electron density of a substituted benzene ring based on the effect that those changes have on the ΔG° for acid equilibrium.

The Hammett Equation

The famous Hammett equation is model used to interpret LFER systems is presented in eq. 4. It is a log-log relationship and involves a plot of the log of the rate or the equilibrium constant for a system against established σ values. Almost all other LFER methods look like this equation.

$$\log \frac{k_X}{k_H} = \rho \sigma \quad (4)$$

The ρ value is the slope of the plot and is called the reaction constant. The σ value is called the substituent parameter.

Let us now construct a Hammett plot for the base hydrolysis of ethyl esters of substituted benzoic acid. It should be identical to figure 1 on page 1 with the exception that the score for electron density is not expressed as σ rather than $-pK_a$. this plot is presented in figure 3.

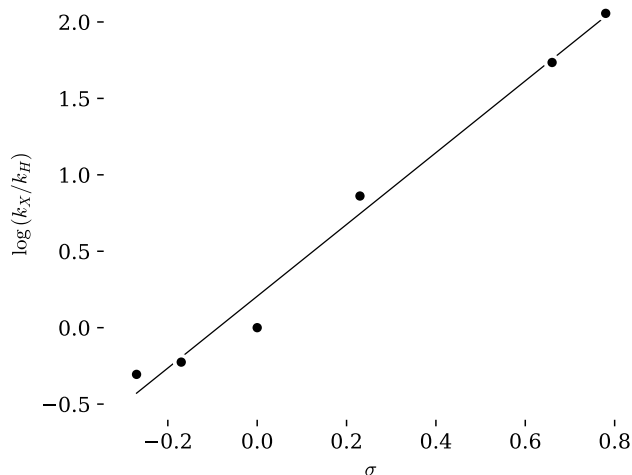


Figure 3: Hammett plot for rate data in the basic hydrolysis of substituted ethyl benzoates.

Line fit equation is...

$$y = (2.35 \pm 0.14)x + (0.21 \pm 0.06)$$

The reaction constant, ρ is 2.4. The data point for unsubstituted ethyl benzoate is at the origin because the σ value for H is zero and $\log k_H/k_H = 0$. The line did not pass through the origin because of scatter in the experimental results. But the reaction constant, as expressed by the slope, is all that really matters.

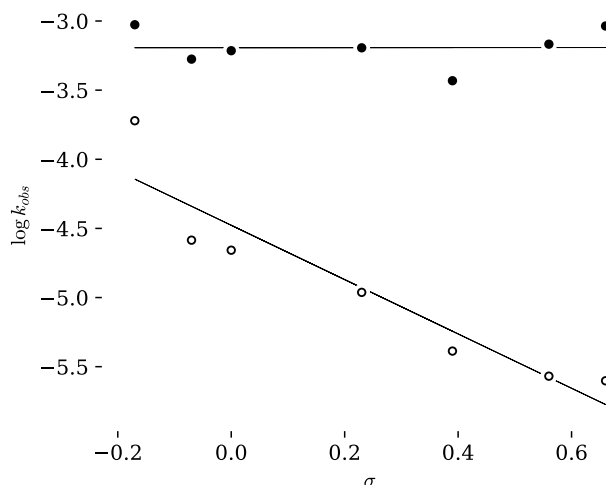
In this section we have answered the first six questions in the handout that were part of the first exercise. Were you able to make a quick plot and evaluate the slope? Did you consider what it meant in context with your reading about the Hammett plot? Next we will continue with more data plots that demonstrate the utility of the Hammett plot.

Exercise: Using the Hammett Equation – 20 minutes

Now we will plot the data used in the second exercise in our class meeting.

Table 4: Benzyl Chloride with Water and with Hydroxide Ion

In table 4 of the class handout we present data for the rate of benzyl chloride reacting with water (in acidic conditions where hydroxide ion concentrations are near zero) and in various concentrations of aqueous sodium hydroxide solution.⁴ The data from table 4 is plotted below. I chose to present the results for the highest concentration of hydroxide and for the acidic reaction.



Both reaction conditions result in S_N2 mechanisms and we expect a transition state structure that reflects a concerted reaction.

⁴ Hydrolysis of benzylchloride in 50% acetone/water in the presence of acid and base. Data from "Relations Entre la Constitution du Substrat et sa Sensibilité à L'ion oh dans L'hydrolyse." S.C.J. Olivier, A.P. Weber, *Recl. Trav. Chim. Pays-Bas*, 1934, 53, 869-890. <https://doi.org/10.1002/recl.19340531002>

Figure 4: Hammett plots for acidic (0.27 M H_2SO_4 , open circles) and basic (0.26 M $NaOH$, closed circles) hydrolysis of benzyl chloride in 50% acetone.

0.26 M $NaOH$, $\rho = -0.00 \pm 0.19$, $R^2 = 0$

0.27 M H_2SO_4 , $\rho = -1.84 \pm 0.32$, $R^2 = 0.94$

This plot is generated from the data table 4 and can be investigated by using the *Python* notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class.14/Table_4_Plots.ipynb.

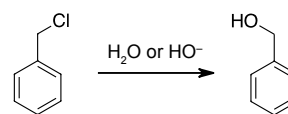
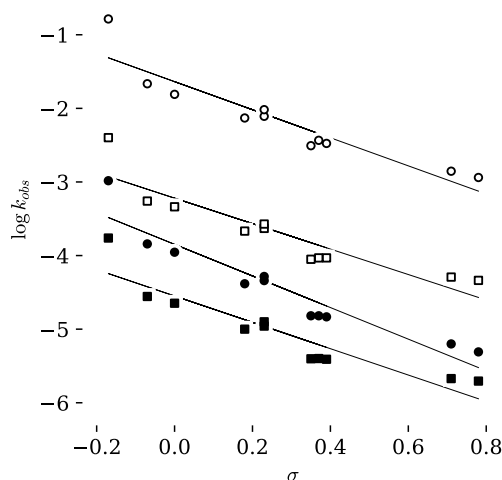


Figure 5: Hydrolysis of benzyl chloride

Table 5: Benzyl Chloride Solvolysis in Water/EtOH

In table 5 of the class handout we present data for the rate of benzyl chloride reacting with solvent in mixtures of water/ethanol or water/acetone.⁵ The data from table 5 is plotted below.



⁵ "Sur un Parallélisme Entre la Mobilité de L'hydrogène du Noyau Benzénique et Celle du Chlore de la Chaîne Latérale" S.C.J. Olivier, *Recl. Trav. Chim. Pays-Bas*, 1930, 49, 697-704. <https://doi.org/10.1002/recl.19300490803>

Figure 6: Hammett plots for solvolysis of benzyl chloride.

Open circles: 48 % EtOH, 83 °C
 $\rho = -1.91 \pm 0.25$, $R^2 = 0.871$

Open squares: 50 % acetone, 60 °C
 $\rho = -1.73 \pm 0.26$, $R^2 = 0.836$

Closed circles: 48 % EtOH, 30 °C
 $\rho = -2.15 \pm 0.25$, $R^2 = 0.893$

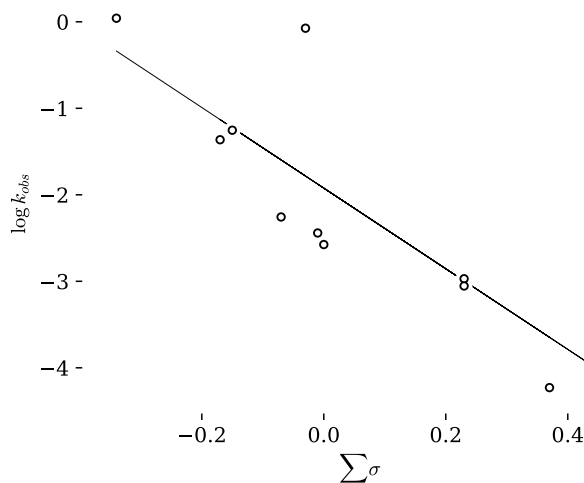
Closed squares: 50 % acetone, 30 °C
 $\rho = -1.79 \pm 0.25$, $R^2 = 0.856$

This plot is generated from the data table 5 and can be investigated by using the *Python* notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_14/Table_5_Plots.ipynb.

Both reaction conditions result in S_N2 mechanisms and we expect a transition state structure that reflects a concerted reaction.

Table 6: Solvolysis of Diphenylmethylene Chloride

In table 6 of the class handout we presented data for the rate of diphenylmethylene chloride reacting with methanol, ethanol and isopropanol at 25 °C.⁶ I chose to plot the data for solvolysis in ethanol.



The negative value for ρ indicates large positive change in charge in the transition state and the large magnitude of the value is consistent with a direct connection to the benzene ring. This indicates a carbocationic intermediate and an S_N1 reaction mechanism.

The poor correlation with Hammett σ values is the result of the carbocationic intermediate. the empty p-orbital is directly connected the the π -system of the benzene ring. *p*-Substituents that have lone pairs can donate electrons directly into the cationic centre and share the charge. this results is a much larger stabilization effect. We need a new set of σ values that reflect this property. In the next meeting we will plot this data again using the Brown-Okamoto σ^+ values.

⁶ S. Altscher, R. Baltzly, S.W. Blackman, *J. Am. Chem. Soc.*, 1952, 74, 3649–3652. <https://doi.org/10.1021/ja01134a054>. J.F. Norris, C. Banta *J. Am. Chem. Soc.*, 1928, 50, 1804–1808. <https://doi.org/10.1021/ja01393a049>. J.F. Norris, J.T. Blake, *J. Am. Chem. Soc.*, 1928, 50, 1808–1812. <https://doi.org/10.1021/ja01393a050>.

Figure 7: Hammett plot for solvolysis of diphenylmethylene chloride. Correlation is poor but we still see the trend. Perhaps we should use different σ parameters?

$$\rho = -4.66 \pm 0.97, R^2 = 0.72$$

This plot is generated from the data table 6 and can be investigated by using the Python notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class.14/Table_6_Plots.ipynb.

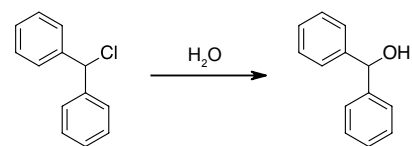
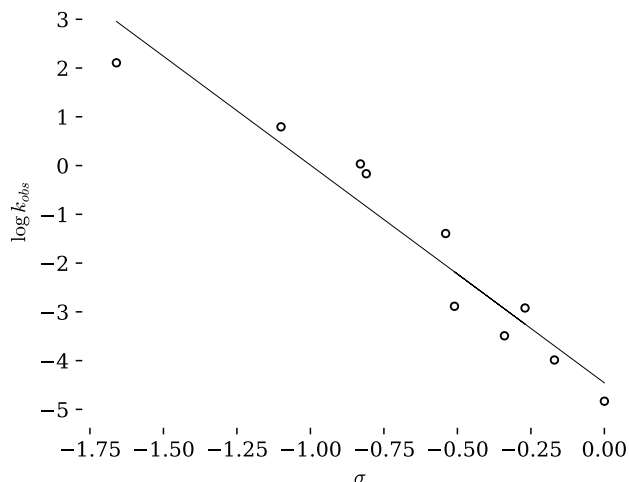


Figure 8: Solvolysis of Diphenylmethylene Chloride

Table 7: Solvolysis of Trityl Chloride

In table 7 of the class handout we presented data for the rate of substituted trityl chlorides reacting with water in acetonitrile 25 °C.⁷



I chose to plot the data for solvolysis in 90% acetonitrile but you can plot other data using the *Python* notebook.

⁷ "Stabilities of Trityl-Protected Substrates: The Wide Mechanistic Spectrum of Trityl Ester Hydrolyses." M. Horn, H. Mayr, *Chem. Eur. J.*, 2010, 16, 7469-7477. <https://doi.org/10.1002/chem.200902669>.

Figure 9: Hammett plot for solvolysis of trityl chloride. Correlation is good with Hammett σ parameters but, as we will see in the next meeting, we should have used Brown-Okamoto σ^+ values

$$\rho = -4.47 \pm 0.44, R^2 = 0.93$$

This plot is generated from the data table 7 and can be investigated by using the *Python* notebook available via Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class.14/Table.7_Plots.ipynb.

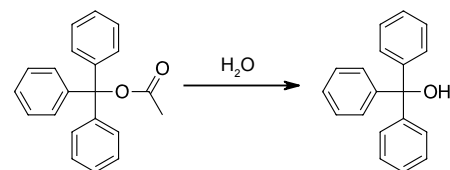


Figure 10: Solvolysis of Trityl Chloride

Analysis & Comments

Hammett plots can provide information about the transition state in a reaction. Let us consider the three cases above.

S_N1 Solvolysis from Data Tables 6 & 7

The plots for the solvolysis of diphenylmethylene chloride and triphenylmethylene chloride (trityl chloride) show that the mechanism involves the formation of a carbocation that is directly attached to the benzene ring. The reaction constants were negative and large in magnitude. In these cases it is more appropriate to use the Brown–Okamoto σ^+ parameters that include the effects of direct resonance stabilization of the cation by some substituents and we will discuss these two data sets in more detail in the following class meeting.

S_N2 Solvolysis Data from Table 4

In S_N2 substitution the transition state is not a carbocation. As electrons depart with the leaving group they are entering with the nucleophile and the electron density at the central carbon remains somewhat constant. There is no carbocation to accept electrons from a resonance donor and so Hammett σ values would be expected to work well here.

From the [plot for Table 4](#) we see that the Hammett reaction constant, ρ , for S_N2 is near zero. No significant charge difference occurs as we move from the starting material to the transition state. This is a nearly perfectly concerted reaction where approach of the nucleophile is synchronous with departure of the leaving group. Below is a More O’Ferrall–Jencks plot that describes this reaction.

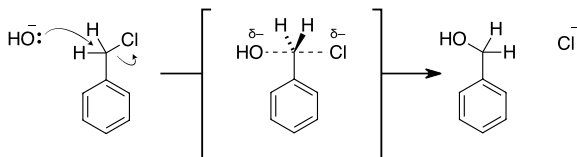


Figure 11: Mechanism of a concerted S_N2 reaction

In the case of water as the nucleophile we observe that $\rho = -1.8$. This indicates that the transition state is more positively charged at the reaction centre compared to the starting material. Obviously, since the intermediate is a positively charged protonated alcohol, we would expect the transition state to have some properties of the intermediate.

However, as the water nucleophile gains positive charge, the chloride leaving group gains negative charge. This would balance the charge in the transition state if the system was perfectly synchronous and we would see a small value for ρ . If we consider how the energy surface of the More O’Ferrall–Jencks plot changes as we change a strong hydroxide nucleophile

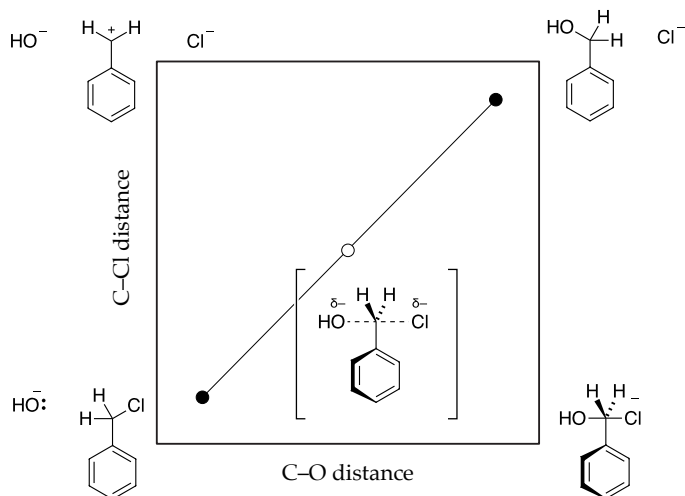


Figure 12: More O'Ferrall-Jencks plot describing S_N2 substitution with hydroxide ion and benzyl chloride.

for a weak water nucleophile, we see that the transition state is no longer positioned to describe a symmetrical structure. The bond to the leaving group is now longer in the t.s. This reduces the electron density at the centre carbon and creates positive charge on the atom directly connected to the benzene ring.

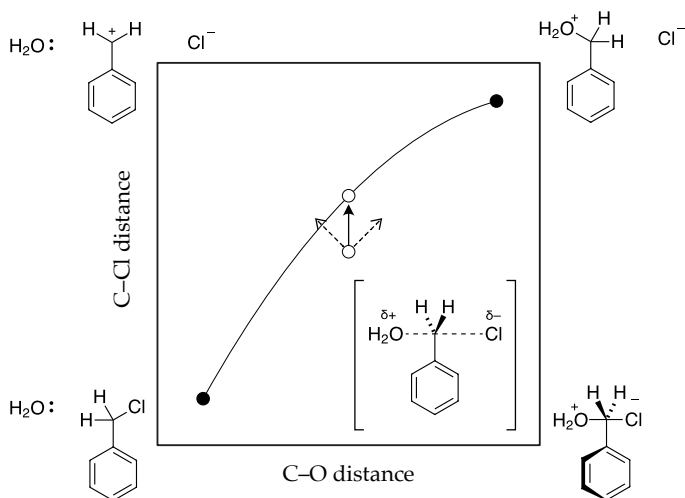


Figure 13: More O'Ferrall-Jencks plot describing S_N2 substitution with water and benzyl chloride.

So the Hammett plot confirms the hypothesis generated by the More O'Ferrall-Jencks plot. The t.s. is not symmetrical

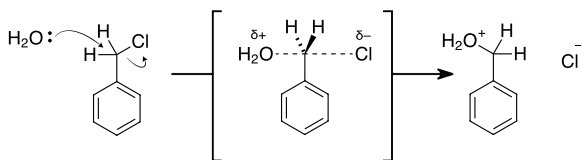


Figure 14: Mechanism of a concerted S_N2 reaction with an asymmetrical t.s.

One could then predict that using a stronger nucleophile than hydroxide would move the position of the t.s. in the opposite direction. In this case the bond to the leaving group would be shorter in the t.s. compared to the hydroxide case. If the leaving group is departing later compared to the approach of the nucleophile then we will be squeezing electrons together at the reaction centre.

One experimental result that I found was for iodide anion acting as a nucleophile. Iodide is a stronger nucleophile than hydroxide ($I^- > Br^- > Cl^-$)⁸ and so we expect to see evidence of a t.s. with increased electron density at the reaction centre. The Hammett plot shows that $\rho = 0.79$.⁹

S_N2 Solvolysis Data from Table 5

The results agree with the case above. the weak nucleophile required significant departure of the good leaving group to assist its' attack. As a result the benzyl carbon is electron-deficient in the transition state. We see that the most electron-donating substituent, –OMe, is off the line in all four experiments? Why? How can we address this?

Conclusions

We see that Hammett plots reveal information about changes in charges in transition states. We can use these tools of test hypotheses about the t.s. for a given reaction.

⁸ Recall that larger halogen ions are more polarizable and are better nucleophiles *and* better leaving groups. Hydroxide is very well solvated in water and is less strong than you might expect.

⁹ You can find the data in "Velocities of reaction of substituted benzyl chlorides in two reactions of opposed polar types." G. M. Bennett, B. Jones, *J. Chem. Soc.*, 1935, 1815–1819. <https://doi.org/10.1039/JR9350001815>. Plot the data for yourself and confirm the stated value for ρ .